

Diurnal Temperature Range Trends in St. John's, NL (2000–2025)

Isaac Adoboe

Section 1

Motivation & Context

Motivation & Context

- **Diurnal Temperature Range (DTR)** = $T_{\max}(t) - T_{\min}(t)$
- DTR reflects atmospheric stability, cloud cover, and air mass characteristics
- **Global pattern:** DTR declining due to asymmetric warming
- **St. John's context:** Maritime climate; winter DTR suppressed, summer DTR elevated
- **Research question:** Is DTR declining in St. John's consistent with global warming theory?

Section 2

Data & Methods

Data & Methods

Data Sources - Four meteorological stations in St. John's region - Daily minimum and maximum temperatures (2000–2025) - Regional mean DTR computed across stations

Statistical Approach - Linear regression: $\text{DTR}_{\text{annual}} \sim \text{Year}$ - Mann-Kendall test: non-parametric test for monotonic trend - Extreme events: High-DTR (>90th percentile), Low-DTR (<10th percentile) - Sensitivity: compare global vs. seasonal threshold definitions - Significance level: $\alpha = 0.05$

Section 3

Exploratory Analysis: Seasonal DTR Pattern